# Banach Fixed-Point Theorem

# An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1]*Mindverse Computing LLC, Lynnwood, WA 98087, USA*

(Dated: October 2026)

Abstract

A complete authored walkthrough of banach fixed-point theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

**RA-03 / Real Analysis / Report v1**

## I. TARGET AND IMPORT CONTRACT

**Theorem I.1.** *Let $(X, d)$ be a nonempty complete metric space and $T : X \to X$ satisfy $d(Tx, Ty) \leq qd(x, y)$ for all $x, y$, with a fixed $0 \leq q < 1$. Then there is a unique $x_*$ with $Tx_* = x_*$. From any $x_0 \in X$, the sequence $x_{n+1} = Tx_n$ converges to $x_*$. For $0 < q < 1$,*

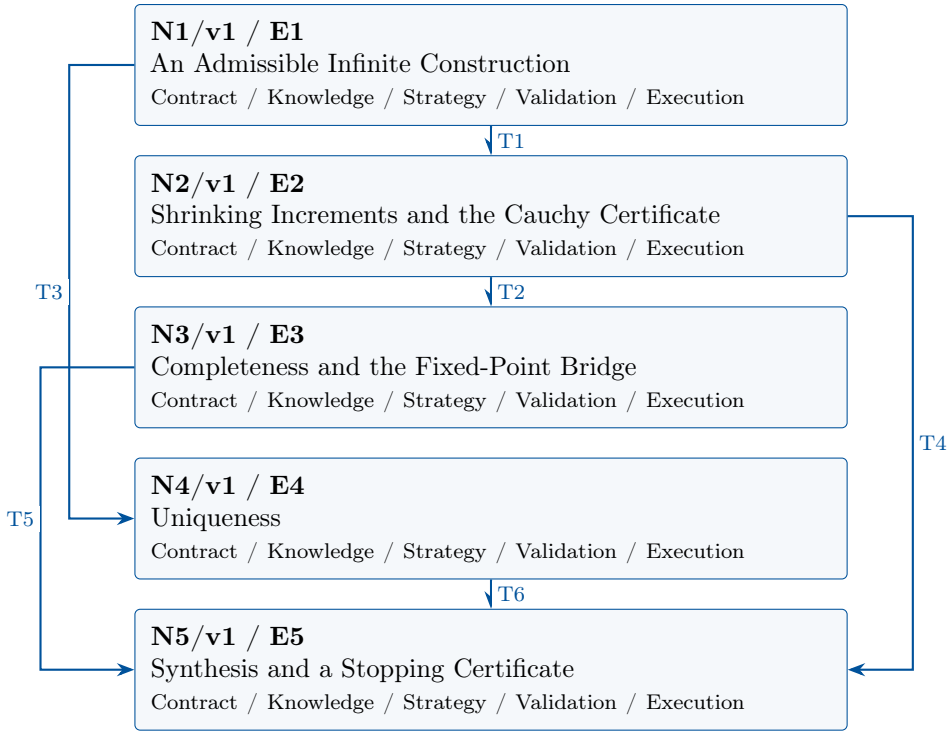
$$d(x_n, x_*) \leq \frac{q^n}{1-q} d(x_1, x_0).$$

*The output is an existence proof, a uniqueness proof, a convergent construction, and an error certificate. An iterate with small consecutive displacement is an approximation; it is not automatically an exact fixed point.*

### A. Foundations and permitted imports

completeness means every Cauchy sequence in X has a limit in X; a contraction has one uniform factor q; geometric-series summation and the triangle inequality are imported. If $q = 0$, T is constant, so $x_1$ is already a fixed point. We prove the remaining case $0 < q < 1$.

## II. NODE INTERACTION WORKFLOW




Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

### A. Versioned evidence-transfer ledger

**T1: N1/E1/v1 to N2.** For every $x_0 \in X$, $x_{n+1} = T(x_n)$ stays in the nonempty complete space $X$.

**Receiving activity.** N2: Shrinking Increments and the Cauchy Certificate

**Compatibility.** Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 and the finite geometric-series bound.

**T2: N2/E2/v1 to N3.** $d(x_{n+p}, x_n) \leq q^n d(x_1, x_0)/(1-q)$; the iterates are Cauchy.

**Receiving activity.** N3: Completeness and the Fixed-Point Bridge

**Compatibility.** Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2 and completeness of the original metric space.

**T3: N1/E1/v1 to N4.** For every $x_0 \in X$, $x_{n+1} = T(x_n)$ stays in the nonempty complete space $X$.

**Receiving activity.** N4: Uniqueness

**Compatibility.** Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1; two putative exact fixed points $u, v$.

**T4: N2/E2/v1 to N5.** $d(x_{n+p}, x_n) \leq q^n d(x_1, x_0)/(1-q)$; the iterates are Cauchy.

**Receiving activity.** N5: Synthesis and a Stopping Certificate

**Compatibility.** Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, N3/E3, and N4/E4.

**T5: N3/E3/v1 to N5.** There exists $x_* \in X$ with $T(x_*) = x_*$ and $d(x_n, x_*) \leq q^n d(x_1, x_0)/(1-q)$.

**Receiving activity.** N5: Synthesis and a Stopping Certificate

**Compatibility.** Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, N3/E3, and N4/E4.

**T6: N4/E4/v1 to N5.** Any two fixed points in $X$ are equal.

**Receiving activity.** N5: Synthesis and a Stopping Certificate

**Compatibility.** Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, N3/E3, and N4/E4.

## III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

### A. N1/v1: An Admissible Infinite Construction

**Object and role.** An Admissible Infinite Construction. Establish the following local result: For every $x_0 \in X$, $x_{n+1} = T(x_n)$ stays in the nonempty complete space $X$.

**Inputs.** $T : X \to X$ and a uniform contraction constant $0 \leq q < 1$.

**C — Contract**

Own E1/v1: For every $x_0 \in X$, $x_{n+1} = T(x_n)$ stays in the nonempty complete space $X$.

**K — Knowledge**

Metric and map definitions

**S — Strategy**

Choose any starting point

**V — Validation**

Check completeness, self-map, $q < 1$

**E — Execution**

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

**Numbered mathematical activities**

N1.1 Check nonemptiness, completeness, the self-map condition, and a single global constant $q$.

N1.2 If $q = 0$, the map is constant and its value is its unique fixed point.

N1.3 For $0 < q < 1$, fix any $x_0$ and form the iteration.

**Export E1/v1.** For every $x_0 \in X$, $x_{n+1} = T(x_n)$ stays in the nonempty complete space $X$.

**Acceptance gate.** Check completeness, self-map, $q < 1$ Publish only after checking every displayed deduction.

**Targeted repair.** If the check fails, revise N1: Check completeness, self-map, $q < 1$ Recheck dependent nodes N2, N3, N4, N5.

## B. N2/v1: Shrinking Increments and the Cauchy Certificate

**Object and role.** Shrinking Increments and the Cauchy Certificate. Establish the following local result: $d(x_{n+p}, x_n) \le q^n d(x_1, x_0)/(1-q)$; the iterates are Cauchy.

**Inputs.** N1/E1 and the finite geometric-series bound.

**C — Contract**

Own E2/v1: $d(x_{n+p}, x_n) \le q^n d(x_1, x_0)/(1-q)$; the iterates are Cauchy.

**K — Knowledge**

Geometric series

**S — Strategy**

Sum shrinking increments

**V — Validation**

Keep q uniform

**E — Execution**

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

**Numbered mathematical activities**

N2.1 Inductively prove $d(x_{n+1}, x_n) \le q^n d(x_1, x_0)$.

N2.2 Sum increments from $n$ to $n+p$ using the triangle inequality.

N2.3 Let $n$ grow while keeping the tail estimate uniform in $p$.

**Export E2/v1.** $d(x_{n+p}, x_n) \le q^n d(x_1, x_0)/(1-q)$; the iterates are Cauchy.

**Acceptance gate.** Keep q uniform Publish only after checking every displayed deduction.

**Targeted repair.** If the check fails, revise N2: Keep q uniform Recheck dependent nodes N3, N5.

### C. N3/v1: Completeness and the Fixed-Point Bridge

**Object and role.** Completeness and the Fixed-Point Bridge. Establish the following local result: There exists $x_* \in X$ with $T(x_*) = x_*$ and $d(x_n, x_*) \le q^n d(x_1, x_0)/(1-q)$.

**Inputs.** N2/E2 and completeness of the original metric space.

**C** **Contract**

Own E3/v1: There exists $x_* \in X$ with $T(x_*) = x_*$ and $d(x_n, x_*) \le q^n d(x_1, x_0)/(1-q)$.

**K** **Knowledge**

Completeness

**S** **Strategy**

Use contraction continuity

**V** **Validation**

Keep limit in X

**E** **Execution**

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

**Numbered mathematical activities**

N3.1 Take the Cauchy limit inside $X$.

N3.2 Use $d(Tx_*, x_*) \le qd(x_*, x_n) + d(x_{n+1}, x_*)$ and pass to the limit.

N3.3 Let the tail endpoint tend to infinity in the estimate.

**Export E3/v1.** There exists $x_* \in X$ with $T(x_*) = x_*$ and $d(x_n, x_*) \le q^n d(x_1, x_0)/(1-q)$.

**Acceptance gate.** Keep limit in X Publish only after checking every displayed deduction.

**Targeted repair.** If the check fails, revise N3: Keep limit in X Recheck dependent nodes N5.

### D. N4/v1: Uniqueness

**Object and role.** Uniqueness. Establish the following local result: Any two fixed points in $X$ are equal.

**Inputs.** N1/E1; two putative exact fixed points $u, v$.

**C — Contract**

Own E4/v1: Any two fixed points in $X$ are equal.

**K — Knowledge**

Distance axioms

**S — Strategy**

Apply contraction to their distance

**V — Validation**

Compare exact fixed points

**E — Execution**

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

**Numbered mathematical activities**

N4.1 Apply the contraction inequality to $u, v$.

N4.2 Obtain $(1 - q)d(u, v) \leq 0$.

N4.3 Use $1 - q > 0$ and the metric separation axiom.

**Export E4/v1.** Any two fixed points in $X$ are equal.

**Acceptance gate.** Compare exact fixed points Publish only after checking every displayed deduction.

**Targeted repair.** If the check fails, revise N4: Compare exact fixed points Recheck dependent nodes N5.

### E. N5/v1: Synthesis and a Stopping Certificate

**Object and role.** Synthesis and a Stopping Certificate. Establish the following local result: The unique fixed point is reached in the limit from every starting point, with $d(x_n, x_*) \le d(x_n, x_{n+1})/(1-q)$.

**Inputs.** N2/E2, N3/E3, and N4/E4.

**C — Contract**

Own E5/v1: The unique fixed point is reached in the limit from every starting point, with $d(x_n, x_*) \le d(x_n, x_{n+1})/(1-q)$.

**K — Knowledge**

All predecessor certificates

**S — Strategy**

Join construction and bounds

**V — Validation**

Separate approximation from equality

**E — Execution**

Carry out the numbered deductions and publish E5/v1 together with its hypotheses and proof location.

**Numbered mathematical activities**

N5.1 Restart the geometric-tail estimate at index $n$.

N5.2 Use the observed next increment to produce a certified stopping tolerance.

N5.3 Combine existence, uniqueness, and both error estimates without equating a finite iterate to the exact limit.

**Export E5/v1.** The unique fixed point is reached in the limit from every starting point, with $d(x_n, x_*) \le d(x_n, x_{n+1})/(1-q)$.

**Acceptance gate.** Separate approximation from equality Publish only after checking every displayed deduction.

**Targeted repair.** If the check fails, revise N5: Separate approximation from equality Withhold this terminal result until its gate passes.

## IV. CONTINUOUS MATHEMATICAL PROOF

### A. N1 — An Admissible Infinite Construction

Choose $x_0 \in X$ and put $x_{n+1} = Tx_n$. Every iterate stays in X because T is a self-map. Set $D = d(x_1, x_0)$. If $D = 0$, the initial point is already fixed. Otherwise the following estimates still apply without changing the algorithm.

### B. N2 — Shrinking Increments and the Cauchy Certificate

Induction using the contraction gives

$$d(x_{n+1}, x_n) \leq q^n D.$$

For $m > n$, the triangle inequality therefore gives

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} q^k D \leq \frac{q^n D}{1-q}.$$

The bound tends to zero independently of m. Given any epsilon, choose n large enough that the last expression is smaller than epsilon. Thus the sequence is Cauchy. This is a statement about every sufficiently late pair of iterates, not just adjacent ones.

### C. N3 — Completeness and the Fixed-Point Bridge

Completeness supplies $x_* \in X$ with $x_n \to x_*$. Since

$$d(Tx_*, x_*) \leq d(Tx_*, Tx_n) + d(x_{n+1}, x_*) \leq qd(x_*, x_n) + d(x_{n+1}, x_*),$$

the right side tends to zero. Hence $Tx_* = x_*$. Letting m tend to infinity in N2's estimate gives the stated error bound. The continuity used here follows from the contraction inequality itself.

### D. N4 — Uniqueness

If x and y are fixed points, then $d(x, y) = d(Tx, Ty) \leq qd(x, y)$. Consequently $(1 - q)d(x, y) \leq 0$. As $1 - q > 0$ and distance is nonnegative, $d(x, y) = 0$, so $x = y$.

### E. N5 — Synthesis and a Stopping Certificate

Every initial point has a limit by N2–N3, and N4 shows all those limits are the same point. For a practical stopping criterion, N2 applied starting at $x_n$ gives

$$d(x_n, x_*) \leq \frac{d(x_{n+1}, x_n)}{1-q}.$$

For $n \geq 1$ there is also the sharper bound

$$d(x_n, x_*) \leq \frac{q}{1-q} d(x_n, x_{n-1}).$$

Thus a requested tolerance can be met using observed increments and a proved contraction factor. This completes the theorem. □

□

## V. WORKED ORCHESTRATION

Take $X = [0, 1]$ with ordinary distance and $T(x) = \cos$ x. The interval is complete, and T maps it into [cos 1,1], a subset of X. The mean value theorem gives $|\cos x - \cos y| \leq (\sin 1)|x - y|$. Use $q = \sin 1 \approx 0.841471 < 1$.

Starting from $x_0 = 0$:

- **n**

  0

  **Approximate iterate**

  0

- **n**

  1

  **Approximate iterate**

  1

- **n**

  2

  **Approximate iterate**

  0.540302306

- **n**

  3

  **Approximate iterate**

  0.857553216

- **n**

  4

  **Approximate iterate**

  0.654289790

- **n**

  5

  **Approximate iterate**

  0.793480359

The alternation does not invalidate convergence. A contraction bound controls its shrinking oscillations. After 100 iterations, the a priori error bound is $q^{100}/(1-q) < 2.1 \times 10^{-7}$. The limit is approximately 0.7390851332; this decimal is illustrative, while the theorem establishes the unique exact solution in [0,1].

### A. Boundary and hypothesis audit

- **Boundary**

  $q = 0$

  **Consequence**

  T is constant; one application reaches the fixed point

- **Boundary**

  $q = 1$

  **Consequence**

  Identity on a multi-point space has many fixed points; uniqueness fails

- **Boundary**

  Missing completeness

  **Consequence**

  Limit may leave the space

- **Boundary**

  Map leaves X

  **Consequence**

  Iteration is not covered by the contract

- **Boundary**

  Finite decimal equality

  **Consequence**

  Does not certify an exact fixed point

Export the contraction theorem, uniqueness lemma, and both error estimates with their hypotheses. The Inverse Function Theorem example imports this completed module to solve nearby equations.

## VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

### A. Failure and correction

Consider $T(x) = x/2$ on $X = (0, 1)$. A proposed branch claims a fixed point because the distances halve. Validation finds that X is not complete: the iterates converge to 0, which is missing.

- **Node:** N3
- **State:** STALLED
- **Failing component:** Validation
- **Failed hypothesis:** complete domain containing the limit
- **Example:** $T(x) = x/2$ on (0,1)
- **Repair target:** N1
- **Blocked export:** fixed-point existence in (0,1)

Knowledge retrieves completeness's definition. Validation retains the missing-point diagnosis. Strategy proposes [0,1] for this example. Execution verifies that the repaired domain is complete and invariant and that 0 is fixed. Contract publishes a theorem for the repaired domain while recording that the original map on (0,1) has no fixed point. Repair does not retroactively prove the rejected original claim.

### B. Hypothesis boundary

- **Boundary**

  $q = 0$

  **Consequence**

  T is constant; one application reaches the fixed point

- **Boundary**

  $q = 1$

  **Consequence**

  Identity on a multi-point space has many fixed points; uniqueness fails

- **Boundary**

  Missing completeness

  **Consequence**

  Limit may leave the space

- **Boundary**

  Map leaves X

  **Consequence**

  Iteration is not covered by the contract

- **Boundary**

  Finite decimal equality

  **Consequence**

  Does not certify an exact fixed point

Export the contraction theorem, uniqueness lemma, and both error estimates with their hypotheses. The Inverse Function Theorem example imports this completed module to solve nearby equations.

### C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

**Provenance.** This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

**Further reading.** University of Pennsylvania advanced-analysis material on contraction-based inversion.